SPACE MISSION DESIGN
AND OPERATIONS

ORBIT NORTH

Design a CubeSat mission. Command a live mission-control simulation. Build a career in space — over one term.

WINTER 2027 · UNIVERSITY OF OTTAWA · 3 CREDITS



12
WEEKS · ONE TERM

1
MISSION · END TO END

2
CODES · UG & GRAD

0
PRIOR SPACE EXPERIENCE NEEDED

> ORBIT NORTH takes you from the Canadian space sector and its regulations, through a complete preliminary CubeSat design in a Concurrent Design Facility, to the seat of a mission-control console — across a twelve-week term at the University of Ottawa. Built for students with the drive to do something real.

YOUR 12 WEEKS

WEEKS 1 – 2
Introduction

The Canadian space landscape and your mission concept

Start with how real missions are built: the Canadian space sector, systems engineering, the concurrent design method and the SpaceCDF platform. Move from a mission need to a problem statement; in the Week 2 studio the design facility opens, and your first concept of operations goes to an informal Mission Concept Review at the end of the week.

CANADIAN SECTOR · SPACECDF · CONOPS · MCR

WEEKS 3 – 7
Design

Concurrent Design Facility

Lock the concept and meet Canadian regulation in Week 3 — the Remote Sensing Space Systems Act, ISED spectrum licensing, export controls and debris mitigation. Then one design stage per week on the SpaceCDF platform, each of you in a discipline role: the requirements baseline; orbit, power, attitude and thermal; communications, structure and propulsion, with two formal trade studies; then integration, verification, cost and risk, ending with a Preliminary Design Review in Week 7.

RSSSA · ISED · REQUIREMENTS · SUBSYSTEM DESIGN · TRADE STUDIES · PDR

WEEKS 8 – 12

Operations

Live mission operations simulation

Two weeks of operations training in a mission control room (Weeks 8–9): positions, the voice loop, pass planning, anomaly response and dry runs. Then two six-hour simulation blocks flying the EOSAT-1 Earth-observation satellite (Weeks 10–11) — launch, early operations and commissioning, then nominal operations with contingencies injected in real time. One team operates while another observes, then the teams swap. Week 12 reviews the whole mission, from the need to the last pass.

MISSION CONTROL · LEOP · ANOMALY RESPONSE · MISSION REVIEW

HOW THE TERM WORKS

1.5 h
Lecture and seminar
Each week, a short lecture on the week's topic, then discussion.

3 h
Studio
Team work in the design facility (Weeks 1–7) or the mission control room (Weeks 8, 9 and 12).

6 h
Simulation blocks
Weeks 10 and 11: a full operations shift, run without a break.

In person, with a reading break after Week 6. Nothing to buy: the open textbook *Concurrent Spacecraft Design*, the course videos, SpaceCDF and the simulator are all provided.

WHAT YOU'LL BE ABLE TO DO

- → Structure a space mission concept using ECSS and NASA frameworks
- → Conduct stakeholder analysis grounded in the Canadian context
- → Execute spacecraft operations procedures in a mission control environment
- → Design a CubeSat mission across all principal subsystems in a Concurrent Design Facility
- → Navigate Canadian satellite licensing, export controls and RSSSA obligations
- → Respond to anomalies and contingency scenarios in a live simulation

TWO CODES · DIFFERENTIATED OUTCOMES

GNG 3100
Topics in Engineering I
Undergraduate. No prior space background required.

SYS 5186
Advanced Topics in Systems Engineering
Graduate. Additional outcomes in quantitative trade-off evaluation, sub-area leadership and literature-grounded synthesis.

WHO'S INVOLVED

University of Ottawa · Faculty of Engineering · SEDTI

INTERESTED? GET IN TOUCH.

Express your interest →

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APPLY